MSD MANUAL
Professional Version

Pulmonary Hypertension

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Pulmonary hypertension is increased pressure in the pulmonary circulation. It has many secondary causes; some cases are idiopathic. Pulmonary vessels may become constricted, pruned, lost, and/or obstructed. Severe pulmonary hypertension leads to right ventricular overload and failure. Symptoms are fatigue, exertional dyspnea, and, occasionally, chest discomfort and syncope. Diagnosis is made by finding elevated pulmonary artery pressure (estimated by echocardiography and confirmed by right heart catheterization). Treatment is with pulmonary vasodilators and diuretics. In some advanced cases, lung transplantation is an option. Prognosis is poor overall if a treatable secondary cause is not found.

There are 3 distinct hemodynamic profiles of pulmonary hypertension (see also table Hemodynamic Profiles of Pulmonary Hypertension):

- Pre-capillary pulmonary hypertension
- Post-capillary pulmonary hypertension
- Combined pre- and post-capillary pulmonary hypertension

TABLE

Hemodynamic Profiles of Pulmonary Hypertension

Type	Mean Pulmonary Artery Pressure at Rest	Pulmonary Artery Wedge Pressure*	Pulmonary Vascular Resistance†
Pre-capillary pulmonary hypertension (eg, chronic lung	> 20 mm Hg	Normal (≤ 15 mm Hg)	Elevated (> 2 Wood units)

Type	Mean Pulmonary Artery Pressure at Rest	Pulmonary Artery Wedge Pressure*	Pulmonary Vascular Resistance†
disorders, venous thromboembolic disease)			
Post-capillary pulmonary hypertension (eg, left heart disease)	> 20 mm Hg	Elevated (> 15 mm Hg)	Normal (< 2 Wood units)
Combined pre-capillary and post-capillary pulmonary hypertension (eg, chronic pre-capillary vascular remodeling that occurs in patients who also have post-capillary pulmonary hypertension due to heart disease)	> 20 mmHg	Elevated (> 15 mm Hg)	Elevated (> 2 Wood units)

* Also called pulmonary artery occlusion pressure.

† Measured by right heart catheterization. Wood unit, also called hybrid resistance unit, is mm Hg × min/L.

Data from Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Respir J* 2023;61(1):2200879. Published 2023 Jan 6. doi:10.1183/13993003.00879-2022

Etiology of Pulmonary Hypertension

Many conditions and medications cause pulmonary hypertension. The most common overall causes of pulmonary hypertension are

- Left heart failure, including diastolic dysfunction

- Parenchymal lung disease with hypoxia

Several other causes of pulmonary hypertension include sleep apnea, systemic rheumatic diseases, and recurrent pulmonary embolism.

Pulmonary hypertension is currently classified into 5 groups (see table Classification of Pulmonary Hypertension) based on a number of pathologic, physiologic, and clinical factors. In the first group (pulmonary arterial hypertension [PAH]), the primary disorder affects the small pulmonary arterioles.

A small number of cases of PAH occur sporadically, unrelated to any identifiable disorder; these cases are termed idiopathic PAH.

Hereditary forms of PAH (autosomal dominant with incomplete penetrance) have been identified; mutations of the following genes have been found:

- Activin-like kinase type 1 receptor (*ALK-1*)
- Bone morphogenetic protein receptor type 2 (*BMPR2*)
- Caveolin 1 (*CAV1*)
- Endoglin (*ENG*)
- Growth differentiation factor 2 (*GDF2*)
- Potassium channel subfamily K member 3 (*KCNK3*)
- Mothers against decapentaplegic homologue 9 (*SMAD9*)
- T-box transcription factor 4 (*TBX4*)

Mutations in *BMPR2* (the gene coding for bone morphogenetic protein receptor type 2) cause 75% of cases of heritable PAH (1). The other mutations are much less common, occurring in about 1% of cases.

In about 20% of cases of hereditary PAH, the causative mutations are unidentified.

A mutation in the eukaryotic translation initiation factor 2 alpha kinase 4 gene (*EIF2AK4*) has been linked to pulmonary veno-occlusive disease, a form of PAH Group 1 (see table Classification of Pulmonary Hypertension) (2, 3).

Certain **medications and toxins** are risk factors for PAH. Those definitely associated with PAH are fenfluramine, dexfenfluramine (discontinued in the United States), aminorex (discontinued in the United States), toxic rapeseed oil, benfluorex (not available in the United States), amphetamines, methamphetamines, and dasatinib. Similarly, other protein kinase inhibitors have also been linked to drug-induced pulmonary hypertension (4). Selective serotonin reuptake inhibitors taken by pregnant women increase the risk of persistent pulmonary hypertension of the newborn. Other medications and substances that are likely or possibly associated with PAH are amphetamine-like drugs, cocaine, phenylpropanolamine (discontinued in the United States), St. John's wort, interferon-alpha, interferon-beta, alkylating agents, bosutinib (only possibly linked to PAH), direct-acting antiviral agents against hepatitis C virus, leflunomide, indirubin (used in some traditional Chinese medicine preparations), and tryptophan (5).

Patients with hereditary causes of hemolytic anemia, such as sickle cell disease, are at high risk of developing pulmonary hypertension (10% of cases based on right heart catheterization criteria) (6, 7). The mechanism is related to intravascular hemolysis and release of cell-free hemoglobin into the plasma, which scavenges nitric oxide, generates reactive oxygen species, and activates the hemostatic system. Other risk factors for pulmonary hypertension in sickle cell disease include iron overload, liver dysfunction, thrombotic disorders, and chronic kidney disease.

TABLE

Classification of Pulmonary Hypertension

Group	Type	Specific Disorders
1	Pulmonary arterial hypertension (PAH)	Disorders associated with PAH: • Congenital heart disorders • Systemic rheumatic disorders • HIV infection • Portal hypertension • Schistosomiasis Drug- and toxin-induced PAH Heritable PAH: • *BMPR2* • *ALK-1*, *CAV-1*, *ENG*, *GDF2*, *KCNK3*, *SMAD9*, *TBX4* • Unknown Idiopathic PAH PAH with long-term response to calcium channel blockers Persistent pulmonary hypertension of the newborn (PPHN) Pulmonary veno-occlusive disease (PVOD) and/or

Group	Type	Specific Disorders
		pulmonary capillary hemangiomatosis (PCH) • Coagulopathic • Drug- and toxin-induced PVOD • Heritable PVOD (*EIF2AK4*) • Immune (systemic rheumatic disorder) mediated • Infectious (cytomegalovirus (CMV) infection, Epstein-Barr virus (EBV) infection, HIV infection, measles)
2	Pulmonary hypertension with left-heart disease	Congenital or acquired left heart inflow or outflow tract obstruction and congenital cardiomyopathies Left heart diastolic dysfunction, including left heart failure with preserved ejection fraction Left heart systolic dysfunction Valvular heart disorders
3	Pulmonary hypertension associated with lung disorders, hypoxemia, or both	Alveolar hypoventilation disorders Chronic exposure to high altitude Chronic obstructive pulmonary disease (COPD) Developmental abnormalities

Group	Type	Specific Disorders
		Interstitial lung disease Lymphangioleiomyomatosis Sleep-disordered breathing Other pulmonary disorders with a mixed restrictive and obstructive pattern
4	Pulmonary hypertension due to pulmonary artery obstructions	Chronic thromboembolic pulmonary hypertension Nonthrombotic pulmonary embolism (eg, due to tumors, parasites, or foreign materials) Thromboembolic obstruction of distal or proximal pulmonary arteries
5	Miscellaneous (unclear or multifactorial mechanisms)	Hematologic disorders: • Chronic hemolytic anemia • Myeloproliferative disorders • Sickle cell disease • Splenectomy Systemic disorders: • Pulmonary Langerhans cell histiocytosis • Sarcoidosis Metabolic disorders: • Gaucher disease • Glycogen storage disease

Group	Type	Specific Disorders
		• Thyroid disorders Other disorders: • Chronic kidney disease • Fibrosing mediastinitis • Pulmonary tumor thrombotic microangiopathy

ALK-1 = activin-like kinase type 1 receptor; BMPR2 = bone morphogenetic protein receptor type 2; CAV1 = caveolin 1; EIF2AK4 = eukaryotic translation initiation factor 2 alpha kinase 4; ENG = endoglin; GDF2 = growth differentiation factor 2; KCNK3 = potassium channel subfamily K member 3; SMAD9 = mothers against decapentaplegic homologue 9; TBX4 = T-box transcription factor 4.

Data from Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension [published correction appears in *Eur Heart J* 2023 Apr 17;44(15):1312. doi: 10.1093/eurheartj/ehad005]. *Eur Heart J* 2022;43(38):3618-3731. doi:10.1093/eurheartj/ehac237

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Pathophysiology of Pulmonary Hypertension

Pathophysiologic mechanisms that cause pulmonary hypertension include

- Increased pulmonary vascular resistance
- Increased pulmonary venous pressure
- Increased pulmonary venous flow due to congenital heart diseases

Increased pulmonary vascular resistance

Increased pulmonary vascular resistance is caused by

- Pathologic vasoconstriction
- Obliteration of the pulmonary vascular bed

Pulmonary hypertension is characterized by variable and sometimes pathologic vasoconstriction and by endothelial and smooth muscle proliferation, hypertrophy, and chronic inflammation, resulting in vascular wall remodeling. Vasoconstriction is thought to be due in part to enhanced activity of thromboxane and endothelin-1 (both vasoconstrictors) and reduced activity of prostacyclin and nitric oxide (both vasodilators).

The increased pulmonary vascular pressure that results from vascular obstruction further injures the endothelium. Injury activates coagulation at the intimal surface, which may worsen the hypertension.

Thrombotic coagulopathy due to platelet dysfunction, increased activity of plasminogen activator inhibitor type 1 and fibrinopeptide A, and decreased tissue plasminogen activator activity may also contribute. Platelets, when stimulated, may also play a key role by secreting substances that increase proliferation of fibroblasts and smooth muscle cells such as platelet-derived growth factor (PDGF), vascular endothelial growth factor (VEGF), and transforming growth factor-beta (TGF-β). Focal coagulation at the endothelial surface should not be confused with chronic thromboembolic pulmonary hypertension, in which pulmonary hypertension is caused by organized pulmonary emboli.

Mutations in the *BMPR2* gene account for most cases of hereditary PAH and also occurs in idiopathic PAH. Aberrant BMPR2 signaling disturbs the TGF-β/BMP balance, favoring a pro-proliferative and anti-apoptotic response in pulmonary artery smooth muscle and endothelial cells. BMPR2 signaling, therefore, has become an increasingly studied target for pulmonary hypertension treatment (1).

Increased pulmonary venous pressure

Increased pulmonary venous pressure is typically caused by disorders that affect the left side of the heart and raise left chamber pressures, which ultimately lead to elevated pressure in the pulmonary veins. Elevated pulmonary venous pressures can cause acute damage to the alveolar-capillary wall and subsequent edema. Persistently high pressures may eventually lead to irreversible thickening of the walls of the alveolar-capillary membrane, decreasing lung diffusion capacity.

Most commonly, pulmonary venous hypertension occurs in left heart failure with preserved ejection fraction (HFpEF), typically in older females who have hypertension and metabolic syndrome. In patients with pulmonary hypertension related to left-sided heart disease, elevated pulmonary vascular resistance ([PVR] defined as the transpulmonary gradient divided by cardiac output) are associated with worse outcomes.

In most patients, pulmonary hypertension eventually leads to right ventricular hypertrophy followed by dilation and right ventricular failure. Right ventricular failure limits cardiac output during exertion.

Increased pulmonary venous blood flow

Increased pulmonary venous blood flow due to congenital heart disease can cause pulmonary hypertension. This can occur in conditions such as atrial septal defects, ventricular septal defects, and patent ductus arteriosus, presumably through the development of characteristic pulmonary vascular lesions. However, the true effect of increased pulmonary blood flow is poorly defined, and increased flow may lead to vascular obstruction only with concomitant pulmonary vascular resistance or a second stimulus.

Pathophysiology reference

1. Cuthbertson I, Morrell NW, Caruso P. *BMPR2* Mutation and Metabolic Reprogramming in Pulmonary Arterial Hypertension. *Circ Res* 2023;132(1):109-126. doi:10.1161/CIRCRESAHA.122.321554

Symptoms and Signs of Pulmonary Hypertension

Progressive exertional dyspnea and easy fatigability occur in almost all patients. Atypical chest discomfort and exertional light-headedness or presyncope may accompany dyspnea and indicate more severe disease. These symptoms are due primarily to insufficient cardiac output caused by right heart failure.

In advanced disease, signs of right heart failure may include right ventricular heave, widely split second heart sound (S2), an accentuated pulmonic component (P2) of S2, a pulmonary ejection click, a right ventricular third heart sound (S3), tricuspid regurgitation murmur, and jugular vein distention, possibly with v-waves. Liver congestion and peripheral edema are common late manifestations.

Pulmonary auscultation is usually normal.

Patients also may have manifestations of causative or associated disorders.

Diagnosis of Pulmonary Hypertension

- Common presenting symptoms: Exertional dyspnea and fatigue
- Initial evaluation: Chest radiograph, ECG, and echocardiography

- Identification of underlying disorder: Pulmonary function testing, ventilation/perfusion scanning or CT angiography, high-resolution CT (HRCT) of the chest, polysomnography, HIV testing, complete blood count, liver tests, and autoantibody testing
- Confirmation of the diagnosis and gauging severity: Pulmonary artery (right heart) catheterization
- Additional studies to determine severity: 6-minute walk distance and plasma levels of N-terminal pro-brain natriuretic peptide (NT-proBNP) or BNP

Pulmonary hypertension is suspected in patients with significant exertional dyspnea who are otherwise relatively healthy and have no history or signs of other disorders known to cause pulmonary symptoms (1).

Patients initially undergo chest radiography, pulmonary function testing, and ECG to identify more common causes of dyspnea, followed by transthoracic Doppler echocardiography to assess right ventricular function and pulmonary artery systolic pressures as well as to detect structural left heart disease that might be causing pulmonary hypertension.

Complete blood count is obtained to document the presence or absence of erythrocytosis, anemia, and thrombocytopenia.

The most common radiographic findings in pulmonary hypertension are enlarged hilar vessels that rapidly prune into the periphery and a right ventricle that fills the anterior airspace on lateral view.

ECG findings include right axis deviation, R > S in V1, S1Q3T3 (suggesting right ventricular hypertrophy), and peaked P waves (suggesting right atrial dilation) in lead II.

Additional tests are obtained as indicated to diagnose secondary causes that are not apparent clinically. These tests can include

- Ventilation/perfusion scanning or CT angiography to detect thromboembolic disease
- HRCT for detailed information about lung parenchymal disorders in patients in whom CT angiography is not done
- Pulmonary function tests (including spirometry, lung volumes, and diffusion capacity for carbon monoxide [DLCO]) to identify obstructive or restrictive lung disease
- Serum autoantibody tests (eg, antinuclear antibodies [ANA], rheumatoid factor [RF], Scl-70 [topoisomerase I], anti-Ro (anti-SSA), anti-ribonucleoprotein [anti-RNP], and anticentromere antibodies) to gather evidence for or against associated autoimmune disorders
- Polysomnography to identify obstructive sleep apnea

Chronic thromboembolic pulmonary hypertension is suggested by CT angiography or ventilation/perfusion (VQ) scan findings and is confirmed by arteriography. CT angiography is useful to evaluate proximal clot and fibrotic encroachment of the vascular lumen. Other tests, such as HIV testing, liver tests, and polysomnography, are done in the appropriate clinical context.

When the initial evaluation suggests a diagnosis of pulmonary hypertension, pulmonary artery catheterization is necessary to measure the following:

- Right atrial pressure
- Right ventricular pressure
- Pulmonary artery pressure
- Pulmonary artery wedge pressure
- Cardiac output
- Left ventricular diastolic pressure

Right-sided oxygen saturation should be measured to exclude left-to-right shunt through an atrial septal defect. Finding a mean pulmonary arterial pressure of > 20 mm Hg, a pulmonary artery wedge pressure ≤ 15 mm Hg, and a pulmonary vascular resistance > 2 Woods units identifies patients with underlying PAH.

Medications that acutely dilate the pulmonary vessels, such as inhaled nitric oxide, IV epoprostenol, or adenosine, are often given during catheterization in patients with idiopathic, hereditary, or drug-induced PAH. Decreasing right-sided pressures in response to these medications may help in the choice of medications for treatment.

Lung biopsy is neither needed nor recommended because of its associated high morbidity and mortality.

Echocardiography findings of right heart systolic dysfunction (eg, tricuspid annular plane systolic excursion) and certain right heart catheterization results (eg, low cardiac output, very high mean pulmonary artery pressures, and high right atrial pressures) indicate that pulmonary hypertension is severe.

Other indicators of severity in pulmonary hypertension are assessed to evaluate prognosis and to help monitor responses to therapy. They include a low 6-minute walk distance and high plasma levels of N-terminal pro-brain natriuretic peptide (NT-pro-BNP) or brain natriuretic peptide (BNP).

Once pulmonary hypertension is diagnosed, the patient's family history should be reviewed to detect possible genetic transmission (eg, premature deaths in otherwise healthy members of the extended family). In familial PAH, genetic counseling is needed to advise mutation carriers of the risk of disease (approximately 20%) and to advocate serial screening with echocardiography (2). Testing for mutations in the *BMPR2* gene in idiopathic PAH can help identify family members at risk. If patients are negative for *BMPR2*, gene testing for *SMAD9*, *KCN3*, and *CAV1* can further help identify family members at risk.

Diagnosis references

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Treatment of Pulmonary Hypertension

- Avoidance of activities that may exacerbate the condition (eg, cigarette smoking, high altitude, pregnancy, use of sympathomimetics)
- Idiopathic and familial pulmonary arterial hypertension: IV epoprostenol; inhaled, oral, subcutaneous, or IV prostacyclin analogs; oral endothelin-receptor antagonists; oral phosphodiesterase 5 inhibitors, oral soluble guanylate cyclase stimulators; oral prostacyclin (IP2) receptor agonists
- Secondary pulmonary arterial hypertension: Treatment of the underlying disorder
- Lung transplantation
- Adjunctive therapy: Supplemental oxygen, diuretics, and/or anticoagulants

Pulmonary arterial hypertension, Group 1

Treatment of pulmonary arterial hypertension (PAH) is rapidly evolving. Medications target 5 aberrant pathways implicated in the development of PAH:

- Endothelin pathway
- Nitric oxide pathway
- Prostacyclin pathway
- Activin receptor signaling pathway
- Inflammatory pathway

The **endothelin pathway** is targeted by bosentan, ambrisentan, and macitentan, which are oral endothelin-receptor antagonists (ERAs).

The **nitric oxide pathway** is targeted by sildenafil, tadalafil, and vardenafil, which are oral phosphodiesterase 5 (PDE5) inhibitors. Riociguat, a soluble guanylate cyclase stimulator, also acts via the nitric oxide pathway.

The **prostacyclin pathway** is targeted by IV epoprostenol, a prostacyclin analog, which improves function and lengthens survival even in patients who are unresponsive to a vasodilator during catheterization (1). Disadvantages are the need for continuous central catheter infusion and frequent, troubling adverse effects, including flushing, diarrhea, and bacteremia due to the indwelling central catheter. Prostacyclin analogs that are inhaled, taken orally, or given subcutaneously or IV (iloprost and treprostinil) are available. Selexipag is an orally bioavailable small molecule that activates the prostacyclin I2 receptor and lowers mortality and morbidity rates (2).

The **activin receptor signaling pathway** has recently been implicated in pulmonary hypertension and is targeted by sotatercept, an activin signaling inhibitor. Dysregulation of this pathway is characterized by elevated levels of serum and pulmonary activin A, activin B, follistatin, follistatin-like 3 and reduced levels of inhibin-alpha in pulmonary arterial hypertension (3). These proteins, which are members of the transforming growth factor beta (TGF-beta) superfamily, have been implicated in pulmonary vascular remodeling and are helpful in predicting mortality in PAH.

BMPR2 is the most common gene mutation in patients with heritable PAH and in idiopathic PAH. The *BMPR2* gene encodes for the BMP protein, which is also a member of the TGF-beta superfamily. Sotatercept helps restore the balance between anti-proliferation (BMP) and pro-proliferation (activin) signaling pathways that is dysregulated in patients with PAH related to BMP signaling. When added to background therapy for pulmonary hypertension, sotatercept reduced pulmonary vascular resistance in a dose-dependent manner. This change was driven by a reduction in mean pulmonary artery pressures rather than pulmonary artery wedge pressure or cardiac output. This finding was consistent among all background therapy (including prostacyclin infusion therapy) subgroups (4). In a phase 3 trial of sotatercept or placebo added to standard of care background therapy, patients who received sotatercept demonstrated significant improvement in the 6-minute walk test at 24 weeks. Patients randomized to sotatercept also demonstrated significant improvement in pulmonary vascular resistance, NT-proBNP levels, quality of life, risk of death, and World Health Organization (WHO) functional class (5).

The **inflammatory pathway** is targeted by seralutinib, a small-molecule kinase inhibitor that treats the pulmonary artery smooth muscle cell hypertrophy and proliferation as well as perivascular inflammation that occurs in PAH. Seralutinib is a potent inhibitor of derived growth factor receptor alpha and beta, colony stimulating factor 1 receptor, and mast/stem cell growth factor receptor kit (6). In a phase 2 trial, when seralutinib was administered to patients with PAH functional class II or III (see table Functional Status of Patients With Pulmonary Hypertension), there was a significant reduction in the pulmonary vascular resistance (7).

TABLE

Functional Status in Patients With Pulmonary Hypertension

Functional Class	Description
I	No limitation of physical activity; ordinary activity does not cause undue dyspnea, fatigue, chest pain or near syncope
II	Slight limitation of physical activity; patients are comfortable at rest, but ordinary physical activity causes some dyspnea, fatigue, chest pain, or near syncope
III	Marked limitation of physical activity; patients are comfortable at rest and less-than-ordinary activity causes dyspnea, fatigue, chest pain, or near syncope
IV	Inability to perform any physical activity without symptoms; dyspnea and/or

Functional Class	Description
	fatigue may be present at rest. Discomfort is increased by any physical activity, Patients have signs of right-heart failure.

Functional classification of pulmonary hypertension modified after the New York Heart Association functional classification according to the World Health Organization.

Data from Humbert M, Kovacs G, Hoeper MM, et al: 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension [published correction appears in Eur Heart J. 2023 Apr 17;44(15):1312. doi: 10.1093/eurheartj/ehad005]. *Eur Heart J* 2022;43(38):3618-3731. doi:10.1093/eurheartj/ehac237

Guidelines for the initial **approach to therapy** recommend vasoactive testing in the catheterization laboratory. If patients are vasoreactive, they should be treated with a calcium channel blocker. Patients who are not vasoreactive should be treated based on their functional class (see table Functional Status of Patients With Pulmonary Hypertension). Patients who are WHO functional class IV with high-risk findings such as chest pain or syncope should be started on combination therapy with an intravenous or subcutaneous prostacyclin plus ambrisentan (ERA) and tadalafil (PDE5) (8). Patients with WHO functional class IV without high-risk features should be started on combination therapy with ambrisentan and tadalafil or macitentan (ERA) and tadalafil. (9, 10).

With rare exception, **combination therapy** is generally preferred in patients with Group 1 and Group 4 pulmonary hypertension and is supported by a growing body of evidence. For example, in a randomized trial comparing the efficacy of monotherapy with oral ambrisentan 10 mg and oral tadalafil 40 mg to combination therapy of these same 2 medications all taken once daily, combination therapy significantly reduced NT-proBNP levels and increased 6-minute walk distances and the percentage of satisfactory clinical responses (11). Also, adverse clinical outcomes (death, hospitalization, disease progression, or poor long-term outcome) were fewer with combination therapy than with monotherapy. This example supports targeting multiple pathways by beginning treatment of PAH with combination therapy. However, phosphodiesterase 5 inhibitors cannot be combined with riociguat (a soluble guanylate cyclase stimulant) because both medication classes increase cyclic guanosine monophosphate (cGMP) levels, and the combination can lead to dangerous hypotension. Patients with severe right heart failure who are at high risk of sudden death may benefit from early therapy with an intravenous or subcutaneous prostacyclin analog-containing combination regimen (typically in combination with a PDE5 inhibitor and an ERA).

The SERAPHIN study demonstrated a significant reduction in morbidity and mortality with the use of macitentan, which could be delivered alone or as background therapy for PAH (primarily phosphodiesterase 5 inhibitors) (12). In the PATENT-1 trial, riociguat increased 6-minute walk distance, decreased pulmonary vascular resistance, and improved functional class when used as sequential combination therapy in patients receiving an endothelin-receptor antagonist or prostacyclin analog (as

well as when used as monotherapy) (13). The FREEDOM-EV study found that adding oral treprostinil to baseline monotherapy with an ERA or PDE5 inhibitor was more effective than placebo in reducing clinical worsening, reducing NT-proBNP, improving functional class, and improving 6-minute walk distance (14).

Morbidity and mortality are lower with selexipag (a prostacyclin pathway agonist) than with placebo when selexipag is combined with a PDE 5 inhibitor, an ERA, or both (15, 16). However, studies suggest that initial triple-therapy combining macitentan, tadalafil, and selexipag does not improve pulmonary vascular resistance or hemodynamics more than initial double therapy with macitentan and tadalafil alone (9). In certain clinical scenarios in patients with idiopathic, heritable, drug- or toxin-induced, or systemic rheumatic disease associated PAH, adding selexipag is recommended for patients receiving dual oral ERA/PDE5 inhibitor therapy (17).

Selected subgroups are sometimes treated differently. Prostacyclin analogs, endothelin-receptor antagonists, and guanylate cyclase stimulators have been studied primarily in idiopathic PAH; however, these medications can be used cautiously (attending to drug metabolism and drug-drug interactions) in patients with PAH due to systemic rheumatic disease, HIV, or portopulmonary hypertension. Vasodilators should be avoided in patients with PAH due to pulmonary veno-occlusive disease due to the risk of catastrophic pulmonary edema (18).

Lung transplantation offers the only hope of cure but has high morbidity because of rejection (bronchiolitis obliterans syndrome) and infection. The 5-year survival rate is 50% (19). Lung transplantation is reserved for patients with class IV disease (defined as dyspnea associated with minimal activity, leading to bed-to-chair limitations) or complex congenital heart disease in whom all therapies have failed and who meet other health criteria to be a transplant candidate.

Adjunctive therapies to treat heart failure, including diuretics, are necessary for many patients. Most patients should receive warfarin unless contraindicated.

Pulmonary hypertension, groups 2 to 5

Primary treatment involves management of the underlying disorder. Patients with **left-sided heart disease** may need surgery for valvular disease. No multicenter trials have demonstrated benefit from using PAH-specific therapies for pulmonary hypertension secondary to left-heart disease. Therefore, the use of these medications is not recommended for group 2 patients. However, the inhaled prostacyclin analog treprostinil has been shown to improve exercise capacity.

Patients with **lung disorders** and hypoxia benefit from supplemental oxygen as well as treatment of the primary disorder. There is no conclusive evidence to support the use of pulmonary vasodilators in COPD (chronic obstructive pulmonary disease). Inhaled treprostinil has been shown to improve exercise capacity in patients with pulmonary hypertension secondary to interstitial lung disease (20). The use of riociguat and the ERAs are generally contraindicated in pulmonary hypertension secondary to interstitial lung disease due to an increase in adverse events in clinical trials (21, 22).

The first-line treatment for patients with severe pulmonary hypertension secondary to **chronic thromboembolic disease** includes surgical intervention with pulmonary thromboendarterectomy.

During cardiopulmonary bypass, an organized endothelialized thrombus is dissected along the pulmonary vasculature in a procedure more complex than acute surgical embolectomy. This procedure cures pulmonary hypertension in a substantial percentage of patients and restores cardiopulmonary function; operative mortality in high volume centers is < 5% (23). Balloon pulmonary angioplasty is another interventional option. This procedure should be done only at expert centers for symptomatic patients who are not eligible for pulmonary endarterectomy. Riociguat has improved exercise capacity and pulmonary vascular resistance in patients who are not surgical candidates or for whom the risk to benefit ratio is too high (13). Macitentan has also demonstrated improvements in pulmonary vascular resistance, 6-minute walk test, and NT-proBNP levels in inoperable patients with chronic thromboembolic pulmonary hypertension (24). Macitentan has also demonstrated safety when used combination with other PAH therapies, including riociguat (25).

Patients with **sickle cell disease** who have pulmonary hypertension are aggressively treated using hydroxyurea, iron chelation, and supplemental oxygen as indicated. In symptomatic patients with elevated pulmonary vascular resistance with normal pulmonary artery wedge pressure confirmed by right heart catheterization (similar to PAH pathophysiology), selective pulmonary vasodilator therapy (with epoprostenol or an endothelin-receptor antagonist) can be considered. Sildenafil increases incidence of painful crises in patients with sickle cell disease and so should be used only if patients have limited vaso-occlusive crises and are being treated with hydroxyurea or transfusion therapy.

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Prognosis for Pulmonary Hypertension

Five-year transplant-free survival for treated patients is about 50% (1). However, some patient registries suggest lower mortality (eg, 10 to 30% at 1 to 3 years in the REVEAL registry [2]), presumably because currently available treatments are superior. Indicators of a poorer prognosis include

- Lack of response to vasodilators
- Hypoxemia
- Reduced overall physical functioning
- Low 6-minute walk distance
- High plasma levels of NT-pro-BNP or BNP
- Echocardiographic indicators of right heart systolic dysfunction (eg, a tricuspid annular plane systolic excursion of < 1.6 cm, dilated right ventricle, flattened interventricular septum with paradoxical septal motion, and pericardial effusion)
- Right heart catheterization showing low cardiac output, very high mean pulmonary artery pressures, and/or high right atrial pressures

Patients with systemic sclerosis, sickle cell disease, or HIV infection with pulmonary arterial hypertension (PAH) have a worse prognosis than those without PAH.

Prognosis references

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Key Points

- **Pulmonary hypertension is classified into 5 groups.**
- **Suspect pulmonary hypertension if patients have dyspnea unexplained by another clinically evident cardiac or pulmonary disorder.**
- **Begin evaluation with chest radiograph, pulmonary function tests, ECG, and transthoracic Doppler echocardiography.**
- **Confirm the diagnosis by right heart catheterization.**

- Treat group 1 by giving combination therapy with vasodilators and, if these are ineffective, consider lung transplantation.
- Consider treating group 3 with inhaled treprostinil.
- Treat group 4 with pulmonary thromboendarterectomy unless the patient is not a candidate for surgery.
- Treat groups 2, 3, and 5 by managing the underlying disorder, treating symptoms, and sometimes using other measures.

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